KPK BOARD TEXTBOOK NOTES

Physics - Class 9

Unit 1: Physical Quantities and Measurement

Complete notes in English, based on *A Textbook of Physics, Grade IX*
Khyber Pakhtunkhwa Textbook Board (KPTB), Peshawar

> **How to use these notes.** Read one section at a time, learn the green definition boxes by heart, and copy every yellow formula box into your notebook. Work through the solved numericals before you try the multiple-choice questions (MCQs) at the end. The last page is a one-page revision sheet for the night before the exam.

What this unit covers

1. What physics is, its branches, and the scientists named in the book
2. Physical quantities - base quantities and derived quantities
3. The International System of Units (SI) - seven base units
4. Derived units built from base units
5. Scientific notation (standard form)
6. Prefixes to powers of ten
7. Measuring instruments - metre rule, vernier callipers, screw gauge, stopwatch and measuring cylinder
8. Significant figures, least count, zero error and the limits of instruments

After studying this unit, you should be able to

- Describe the crucial role of physics in science, technology and society.
- Explain, with examples, that science is based on physical quantities, which consist of a numerical magnitude and a unit.
- Differentiate between base and derived physical quantities.
- List the seven SI units with their symbols.
- Interconvert prefixes and their symbols for multiples and sub-multiples of units.
- Write answers in scientific notation in measurements and calculations.
- Describe the working of vernier callipers and a screw gauge for measuring length.
- Identify and explain the limitations of measuring instruments such as the metre rule, vernier callipers and screw gauge.
- Describe the need for significant figures when recording and stating laboratory results.

This checklist is the Student Learning Outcomes list printed at the start of Unit 1 in the textbook.

1.1 Introduction to Physics

> **Physics** is the branch of science that involves the study of the physical universe: energy, matter, and how they are related.

People have always been curious about the *how* and *why* behind the working of things. How did the universe begin? Why do things fall to the ground? Why does the Moon seem to change shape during the month? Physics tries to answer questions like these. It covers a wide range of phenomena, from the smallest sub-atomic particles to the largest galaxies and the universe itself.

Mathematics is the language of physics. Physics improved greatly when mathematics was incorporated into it about 400 years ago. Ideas expressed in mathematical terms are clearer, they do not carry the multiple meanings that confuse ordinary discussion, and equations give compact statements of the relationships between concepts. When things in nature are expressed mathematically, they are easier to test by experiment.

Physics and science

Physics is at the root of every field of science. Most of the major developments in chemistry, biology, geology, agricultural science, environmental science, astronomy, engineering and even medicine have been made by physicists.

Physics, technology and society

Physics and technology are closely related, but they are not the same thing. Physics is concerned with gathering knowledge and organising it. Technology lets humans use that knowledge for practical purposes. A physical phenomenon lies behind every technology, so physics has a key role in the progress of humankind and in improving the quality of living.

- **Medicine:** physics provides the basic understanding for instruments such as Computed Tomography (CT scan), Magnetic Resonance Imaging (MRI), Positron Emission Tomography (PET) and laser surgery.
- **Communication:** the use of physics in information technology has improved the standard of communication; mobile phones are now used by everyone, and hologram technology is also incorporated in cell phones.
- **Daily life:** measurement is not only a concern of physics. When we go to a tailor's shop, he takes the required measurements before sewing our clothes. A car cannot be driven safely without its measuring dials.

Contribution of the Islamic world to physical science

Scientist	Contribution given in the textbook
Yaqub Kindi (800-873)	Born in Basra, Iraq. He produced extensive research monographs on metrology, specific gravity and tides. His most important work was in optics, especially the reflection of light.
Ibnal Haitham (965-1039)	Born in Basra, Iraq. His greatest work is the book on optics named *Kitab ul Manazir*. He is also considered the inventor of the pin-hole camera.
Al Beruni (973-1048)	He wrote about 150 books on physics, cosmology, geography, culture, archaeology and medicine. He discussed the shape of the Earth, the movement of the Sun and Moon, and the phases of the Moon.

Famous Pakistani physicists

Scientist	Contribution given in the textbook
Mohammad Abdus Salam (1926-1996)	Pakistani theoretical physicist. He shared the 1979 Nobel Prize in Physics with Sheldon Glashow and Steven Weinberg for his contribution to the electroweak unification theory. He was the first Pakistani to receive a Nobel Prize in science.
Abdul Qadeer Khan (born 1936)	Pakistani nuclear physicist and metallurgical engineer who founded the uranium enrichment programme. He established Kahuta Research Laboratories (KRL) in 1976 and served as its Senior Scientist and Director-General until he retired in 2001.

Branches of Physics

Because the scope of physics and the research in it are so vast, physics is usually divided into a number of branches. The table below follows Table 1.1 of the textbook.

Branch	What it studies	Examples
Mechanics	Motion of objects and how it relates to forces	Falling objects, friction, weight, spinning objects
Heat and Thermodynamics	Heat and temperature and their relation to energy	Melting and freezing, engines, refrigerators
Oscillations and Waves	To-and-fro motion	Springs, water waves, sound
Optics	Light and the instruments created to use or detect it	Mirrors, lenses, telescopes, the eye
Electricity and Magnetism	Static and moving charges and the physical phenomena associated with them	Electric charge, circuitry, magnets, electromagnets
Atomic and Nuclear Physics	Structure and properties of individual atoms and the nuclei of atoms	X-rays, lasers, nuclear reactor, MRI, CT scan, PET scan
Relativity	Moving objects (including at very high speed) and gravitation	Particle accelerators, nuclear energy
Quantum Physics	Discrete, indivisible units of energy called quanta	The atom and its parts
Particle Physics	The nature of the particles that constitute matter and radiation	Quarks, leptons, photons, bosons
Cosmology and Astrophysics	The origin, evolution and eventual fate of the universe	Stars, galaxies, black holes
Biophysics and Medical Physics	Physical interactions in biological processes, and the application of physics to health processes such as prevention, diagnosis and treatment	MRI, CT scan, radiotracers, conduction in living cells

> **Activity 1.1 - Physics in a car.** A single car uses many branches at once: the battery and starter (electricity), the engine and coolant (heat), the spinning wheels and tyres (mechanics), and the rear-view mirrors (optics). When a question asks where physics is used in daily life, the car is a safe, complete example.

1.2 Physical Quantities

Measurement is a comparison between an unknown quantity (a measurable quantity such as mass or length) and a standard, to see how many times as big it is. A **unit** is the standard with which things are compared.

Before a measurement can be made, a standard or unit must be chosen. The size of the quantity to be measured is then found with an instrument that has a scale marked in that unit.

Physical quantities are all the quantities that can be measured. In physics we study only those quantities which can be defined and measured.

Length, mass, time, density and temperature can all be measured, so they are physical quantities. The length of this book can be measured, and the duration of our stay at school is measurable. A physical quantity is always expressed as a **number (magnitude) together with a unit** - for example, a length of 5 metres has the magnitude 5 and the unit metre.

Physical quantities are classified into two categories: **base quantities** and **derived quantities**, explained below.

Base quantities. The minimum number of physical quantities selected, whose units are defined and standardised in such a way that all other physical quantities can be expressed in terms of them. Their units are called **base units**.

Derived quantities. The physical quantities defined in terms of base quantities. Their units are called **derived units**, because they are derived from multiplying and dividing base units.

Base Quantities and Derived Quantities

Point	Base quantity	Derived quantity
Definition	Independent; not defined in terms of any other quantity	Defined in terms of base quantities
Number	Only seven in SI	Very many
Examples	Length, mass, time	Area, volume, speed, force, pressure
Units	metre, kilogram, second	m^2, m^3, m s^{-1}, newton, pascal

1.3 International System of Units (SI)

A complete set of units for all physical quantities is called a **system of units**. The international system of units is abbreviated as **SI**, from the French name *Systeme International d'Unites*, which means the International System of Units. It is based on seven base units, from which all other units are derived.

The seven SI base units (Table 1.2 of the textbook)

Base quantity	Symbol of quantity	SI base unit	Unit symbol
Length	l, x, r, etc.	metre	m
Mass	m	kilogram	kg
Time, duration	t	second	s
Electric current	I or i	ampere	A
Thermodynamic temperature	T	kelvin	K
Amount of substance	n	mole	mol
Luminous intensity	I	candela	cd

In SI, seven physical quantities are chosen as base and their units are defined and standardised; these are called base units. Each SI unit is defined carefully so that it is unique, and so that accurate and reproducible measurements can be made with it.

> **Why SI was needed.** Before SI units were standardised, the inch used as a unit of length in different cities of England had different lengths, and people had to deal with awkward conversion factors such as 16 ounces in a pound, 12 inches in a foot or 5,280 feet in a mile. SI uses decimal notation instead: 1,000 metres is 1 kilometre, and 100 centimetres is 1 metre.

Derived Units

Units derived from multiplying and dividing base units are called derived units. In SI, the units for all other physical quantities can be derived from the seven base units. The unit of velocity is m/s and the unit of acceleration is m/s^2. Some derived units are used so often that they have been given their own special names and symbols: the unit of force, kg m s^{-2}, is given the special name **newton** and the symbol **N**.

Derived units (Table 1.3 of the textbook)

Derived quantity	Symbol	Derived unit	Unit symbol
Area	A	square metre	m^2
Volume	V	cubic metre	m^3
Speed, velocity	v	metre per second	$m\ s^{-1}$
Acceleration	a	metre per second squared	$m\ s^{-2}$
Density	ρ	kilogram per cubic metre	$kg\ m^{-3}$
Force	F	newton	$N = kg\ m\ s^{-2}$
Pressure	P	pascal	$Pa = kg\ m^{-1}\ s^{-2}$
Energy	E, U	joule	$J = kg\ m^2\ s^{-2}$
Capacitance	C	farad	$F = kg^{-1}\ m^{-2}\ s^4\ A^2$
Resistance	R	ohm	$\Omega = kg\ m^2\ s^{-3}\ A^{-2}$

How to derive a unit - worked pattern. Speed = distance / time, so its unit is metre / second = m s^{-1}. Density = mass / volume, so its unit is kilogram / cubic metre = kg m^{-3}. Force = mass x acceleration, so its unit is kg x m s^{-2} = kg m s^{-2}, named the newton (N). Use this same pattern whenever an exam asks you to derive a unit.

1.4 Scientific Notation (Standard Form)

Scientific notation is a way of writing numbers that are too big or too small to be written easily in decimal form. It is commonly used by scientists, mathematicians and engineers. A number is written as the product of a number greater than 1 and less than 10, called the **mantissa**, and a power of 10, called the **exponent**.

$$\text{Number (N)} = \text{Mantissa (M)} \times 10^{\text{exponent (n)}}$$

The mass of the Moon is approximately 70,000,000,000,000,000,000,000 kilograms. In scientific notation this becomes **7×10^{22} kg**, where 7 is the mantissa and 22 is the exponent. The diameter of an atomic nucleus is about 0.00000000000001 m, which in standard form is **1×10^{-14} m**.

Steps for converting a number to standard form

1. Move the decimal point and place it after the first non-zero digit; this makes the mantissa (M).
2. If the decimal point is moved towards the **left**, the power of 10 is **positive**, and the number of digits moved gives the value of the exponent.
3. If the decimal point is moved towards the **right**, the power of 10 is **negative**, and the number of digits moved gives the value of the exponent.

Worked Examples - Scientific Notation

Example 1.1 **Earth-Moon distance.** The average distance between the Earth and the Moon is 384,400,000 m. Write it in standard form.

Solution. To get a mantissa whose first digit before the decimal is non-zero, move the decimal point 8 digits to the left. The power of 10 is therefore positive 8.

$$\text{Distance} = 3.844 \times 10^{8} \text{ m}$$

Example **Bond length.** The bond length of diatomic hydrogen is 0.000000000074 m. Write it in standard form.

Solution. Move the decimal point 11 digits to the right so that it sits after the first non-zero digit (7). The power of 10 is therefore negative 11.

$$\text{Bond length} = 7.4 \times 10^{-11} \text{ m}$$

Assignment 1.3 **Mass of a housefly.** An adult housefly (*Musca domestica*) has a mass of only about 0.0000214 kg. Express it in standard form.

Solution. Move the decimal point 5 digits to the right: mass = **2.14×10^{-5} kg**.

1.5 Prefixes to Powers of Ten

A mechanism through which a very small or a very large number is expressed in terms of a power of ten, by giving a proper name to that power of ten, is called a **prefix** to the power of ten. The thickness of a sheet of paper is written conveniently in millimetres instead of metres, while the long distance between two cities is expressed better in kilometres.

Prefixes (Table 1.5 of the textbook)

Prefix	Multiplier	Symbol	Prefix	Multiplier	Symbol
Exa	10^{18}	E	deci	10^{-1}	d
Peta	10^{15}	P	centi	10^{-2}	c
Tera	10^{12}	T	milli	10^{-3}	m
Giga	10^{9}	G	micro	10^{-6}	μ
Mega	10^{6}	M	nano	10^{-9}	n
kilo	10^{3}	k	pico	10^{-12}	p
hecto	10^{2}	h	femto	10^{-15}	f
deca	10^{1}	da	atto	10^{-18}	a

Exam traps: the largest prefix in this table is **exa** (10^{18}) and the smallest is **atto** (10^{-18}). Note the symbols carefully - capital M is mega, small m is milli; capital T is tera, and deca is written da.

Interconverting Prefixes - Worked Conversions

To convert, replace the prefix by its power of ten, then use scientific notation. Practise each line until the method becomes automatic.

Convert	Working	Answer
5 km into m	5×10^{3} m	5,000 m
250 cm into m	250×10^{-2} m	2.50 m
10 cm into mm	10 cm x 10 mm/cm	100 mm
0.2 mm into m	0.2×10^{-3} m	0.0002 m = 2×10^{-4} m
3.6 Mg into kg	3.6×10^{6} g = 3.6×10^{3} kg	3,600 kg
750 nm into m	750×10^{-9} m	7.5×10^{-7} m

> **Choosing a convenient unit.** Width of a book - centimetre. Length of a room - metre. Diameter of a wire - millimetre. Mass of a candy - gram. Mass of a cricket ball - gram (about 160 g). Always choose the unit that gives a simple number, neither thousands nor long decimals.

1.6 Measuring Instruments

Different quantities need different instruments, and every instrument has a smallest measurement it can make reliably. That smallest measurement is its **least count**.

Instrument	Measures	Typical least count / range in the textbook
Metre rule	Length	1 mm (0.1 cm)
Vernier callipers	Small length, thickness, internal and external diameter	0.1 mm (0.01 cm)
Screw gauge (micrometer)	Very small length, diameter of a wire, thickness of a sheet	0.01 mm
Stopwatch	Time interval	Analog about 0.1-0.2 s; digital 0.01 s
Measuring cylinder	Volume of a liquid, and volume of an irregular solid by displacement	Scale in millilitres (mL) or cubic centimetres (cm^3)
Physical balance / beam balance	Mass	Depends on the smallest standard mass used

Measuring cylinder - volume of an irregular solid

A measuring cylinder is made of transparent plastic or glass and has a vertical scale in millilitres (mL) or cubic centimetres (cm^3). Pour water into it until it is about half full and record the volume (first reading). Lower the irregular solid, such as a key, gently into the water until it is completely immersed, and record the volume again (second reading).

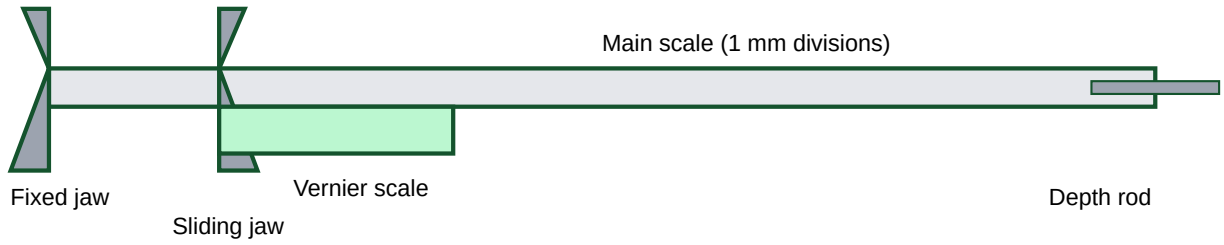
$$\text{Volume of solid} = \text{Second reading} - \text{First reading}$$

Example 1.6 of the textbook, in brief: 1 litre of milk is the volume of a cube of side 10 cm, so its volume is $V = l^3 = (10\ cm)^3 = 10^3\ cm^3 = 1{,}000\ cm^3$. In cubic metres, since $1\ cm = 10^{-2}\ m$, this is $10^3 \times (10^{-2})^3 = 10^{-3}\ m^3$.

1.6.2 Vernier Callipers

> **Vernier callipers** are a device used to measure a fraction of the smallest scale division by sliding another scale over it. They measure a small length accurately up to 0.1 mm or 0.01 cm.

They can be used to measure the thickness, diameter or width of an object, and the internal and external diameter of a hollow cylinder.



The two scales

- **Main (fixed) scale:** it has markings usually of 1 mm each, and it carries the fixed jaw at its left end.
- **Vernier (sliding) scale:** it slides over the main scale. It usually has a length of 9 mm divided equally into 10 divisions, so the separation between two lines on the vernier scale is 9/10 mm = 0.9 mm.

Vernier constant (least count)

The minimum length that can be measured accurately with vernier callipers is called the vernier constant, or least count.

$$\text{Least Count} = 1 \text{ Main Scale Division} - 1 \text{ Vernier Scale Division} = 1 \text{ mm} - 0.9 \text{ mm} = 0.1 \text{ mm} = 0.01 \text{ cm}$$

How to take a reading

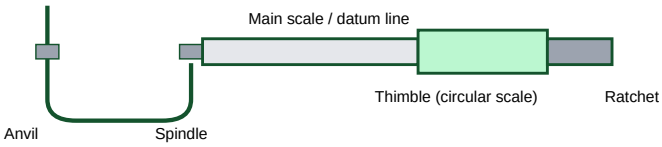
1. Close the jaws gently and check the zero error (see next page).
2. Place the object between the jaws and close them until they just grip it.
3. Read the main scale just before the vernier zero - this is the main scale reading.
4. Find the vernier division that lines up exactly with a main scale division, and multiply it by the least count.
5. Add the two: **Length = Main scale reading + (Coinciding vernier division x Least count)**, then correct for zero error.

Worked Main scale reading = 2.3 cm. The 6th vernier division coincides. Vernier contribution = 6 x 0.01 cm = 0.06 cm. No zero error.

$$\text{Length} = 2.3 \text{ cm} + 0.06 \text{ cm} = 2.36 \text{ cm}$$

1.6.3 Screw Gauge (Micrometer Screw Gauge)

Screw gauge is a device used to measure a fraction of the smallest scale division by the rotatory motion of a circular scale over it. It measures length - for example the diameter of a thin wire or the thickness of a sheet.



Pitch and least count

The **pitch** of a screw gauge is the distance travelled by the circular scale on the linear scale in one complete rotation.

$$\text{Least Count} = \text{Pitch} / \text{Total number of divisions on the circular scale}$$

Textbook example: if the pitch is 0.5 mm and the number of divisions on the circular scale is 50, then the least count = 0.5 mm / 50 = **0.01 mm**.

How to take a reading

1. Close the anvil and spindle gently using the ratchet and check the zero error.
2. Place the wire or sheet between the anvil and the spindle. Turn the ratchet, not the thimble, until it slips - this avoids pressing too hard.
3. Read the main scale reading on the datum line.
4. Read the circular scale division that coincides with the datum line, and multiply it by the least count.
5. **Diameter = Main scale reading + (Coinciding circular division x Least count)**, then correct for zero error.

Worked Pitch = 0.5 mm, circular divisions = 50, so least count = 0.01 mm. Main scale reading = 3.5 mm. The 28th circular division coincides: 28 x 0.01 mm = 0.28 mm. No zero error.

$$\text{Diameter} = 3.5\ \text{mm} + 0.28\ \text{mm} = 3.78\ \text{mm}$$

Zero error - both instruments

When the jaws (or the anvil and spindle) are fully closed, the instrument should read exactly zero. If it does not, it has a **zero error**.

- **Positive zero error:** the closed instrument reads a small positive value. Subtract it from every observed reading.
- **Negative zero error:** the closed instrument reads below zero. Add its magnitude to every observed reading.

$$\text{Correct reading} = \text{Observed reading} - \text{Zero error (with its sign)}$$

Significant Figures

Significant figures are the digits in a measured value that are known reliably, plus the first uncertain digit. A result must not claim more accuracy than the instrument can give: a length measured with a metre rule (least count 1 mm) may be written 25.4 cm, but 25.437 cm pretends to an accuracy the rule does not have.

Rules for counting significant figures

1. All non-zero digits are significant. (5174 has 4 significant figures.)
2. Zeros between non-zero digits are significant. (900.8 has 4 significant figures.)
3. Leading zeros are not significant - they only fix the decimal point. (0.0002 has 1 significant figure.)
4. Trailing zeros after a decimal point are significant. (5174.00 has 6 significant figures; 6.0 has 2.)
5. In scientific notation, count only the digits of the mantissa. (9.460×10^{15} m has 4 significant figures; 6×10^{2} has 1.)

Number	Significant figures	Why
1.1	2	Both digits are non-zero
6.0	2	Trailing zero after the decimal counts
7.1	2	Both digits are non-zero
6×10^{2}	1	Only the mantissa counts
900.8	4	Zeros between non-zero digits count
5174.00	6	Trailing zeros after the decimal count
0.0002	1	Leading zeros do not count
9.460×10^{15}	4	Mantissa 9.460 - the trailing zero counts

These are the exact numbers used in the textbook's exercise MCQs 6, 7 and 8. A light year is about 9.460×10^{15} m - 4 significant figures.

Rounding in calculations

- For multiplication and division, the answer should have no more significant figures than the least precise measurement used.
- If the digit to be dropped is 5 or more, round the last kept digit up; if it is less than 5, leave the last kept digit unchanged.
- Never write extra digits just because the calculator shows them.

Limitations of Measuring Instruments

- **Least count limit:** no instrument can measure smaller than its least count. A metre rule cannot reliably measure 0.05 mm, and vernier callipers cannot measure the thickness of a single hair accurately.
- **Zero error:** worn or misaligned instruments may not read zero when closed, so every reading carries that error unless it is corrected.
- **Parallax error:** reading a scale from an angle instead of square-on gives a wrong value. Keep your eye directly in line with the mark.
- **Reaction time:** a hand-operated stopwatch is limited by human reaction time, about a tenth of a second or more, however fine its scale is.
- **Damaged scales and end error:** a worn end of a metre rule gives wrong readings; start measuring from a clear mark such as 1 cm and subtract it.

Solved Numericals - Practice Set

N1 **Prefix conversion.** Express 0.2 mm in metres.

0.2 mm = 0.2 x 10^{-3} m = 0.0002 m = **2 x 10^{-4} m**. (Both forms are correct - this is MCQ 9 of the textbook exercise.)

N2 **Vernier least count.** A vernier scale of length 9 mm is divided into 10 divisions. Find the least count.

1 vernier division = 9/10 mm = 0.9 mm. Least count = 1 mm - 0.9 mm = **0.1 mm = 0.01 cm**.

N3 **Screw gauge least count.** The pitch of a screw gauge is 1 mm and its circular scale has 100 divisions. Find the least count.

Least count = 1 mm / 100 = **0.01 mm**.

N4 **Zero error correction.** Vernier callipers show a zero error of +0.03 cm. The observed diameter of a ball is 1.54 cm. Find the correct diameter.

Correct reading = 1.54 cm - 0.03 cm = **1.51 cm**.

N5 **Volume by displacement.** A measuring cylinder shows 50 cm^3 of water. After a key is fully immersed, it shows 57 cm^3. Find the volume of the key.

Volume = 57 cm^3 - 50 cm^3 = **7 cm^3**.

N6 **Scientific notation.** Write 0.000,021,4 kg (the mass of a housefly) in standard form.

Move the decimal 5 places to the right: **2.14 x 10^{-5} kg**.

Short Questions - Model Answers

Q1. Why is physics called the root of every field of science?
Because most major developments in chemistry, biology, geology, agriculture, environmental science, astronomy, engineering and medicine have been made by physicists, and every other science uses the quantities, instruments and laws that physics provides.

Q2. Differentiate between base and derived quantities.
Base quantities are the seven independent quantities whose units are defined and standardised so that all other quantities can be expressed in terms of them. Derived quantities are defined in terms of base quantities, and their units are obtained by multiplying and dividing base units.

Q3. Why do we use significant figures in the laboratory?
So that a recorded result claims only the accuracy the instrument can give. Significant figures show which digits are reliable and stop us from writing calculator digits that the measurement never justified.

Exercise MCQs - Textbook (with Answer Key)

1. How many millimetres are there in 10 cm? A. 100 mm B. 200 mm C. 50 mm D. 10 mm
2. Which of the following quantities can be measured using a micrometer? A. current B. force C. length D. mass
3. The instrument that best measures the internal diameter of a pipe is: A. screw gauge B. vernier caliper C. metre rule D. measuring tape
4. Which prefix represents the largest value? A. mega B. giga C. peta D. exa
5. Which of the following is the smallest prefix? A. atto B. pico C. nano D. femto
6. Which number shows one significant digit? A. 1.1 B. 6.0 C. 7.1 D. 6×10^2
7. Which number shows 4 significant digits? A. 900.8 B. 4 C. 5174.00 D. 0.0002
8. A light year is about 9.460×10^{15} m. How many significant figures are in this number? A. 6 B. 2 C. 3 D. 4
9. 0.2 mm in units of metres is: A. 0.0002 m B. 2×10^{-4} m C. none D. both A and B
10. *Kitab ul Manazir* is the name of the book written by: A. Yaqub Kindi B. Ibnal Haitham C. Al Beruni D. none

> **Answer key:** 1-A 2-C 3-B 4-D 5-A 6-D 7-A 8-D 9-D 10-B

One-Page Revision - The Whole Unit

Must-know item	Remember this
Physics	Branch of science studying matter, energy and how they are related
Physical quantity	Any quantity that can be measured = magnitude + unit
Seven base units	m, kg, s, A, K, mol, cd
Derived unit pattern	Multiply / divide base units: speed m s^{-1}, force N = kg m s^{-2}
Scientific notation	$N = M \times 10^n$, 1 <= M < 10; left move = positive power, right move = negative
Biggest / smallest prefix	Exa 10^{18} (E) / Atto 10^{-18} (a)
Vernier least count	0.1 mm = 0.01 cm (1 MSD - 1 VSD)
Screw gauge least count	Pitch / circular divisions; 0.5 mm / 50 = 0.01 mm
Reading formula	Main scale reading + (coinciding division x least count) - zero error
Volume of irregular solid	Second cylinder reading - first cylinder reading
Significant figures	Non-zero digits always count; leading zeros never; zeros between digits and trailing zeros after a decimal do
Book to remember	*Kitab ul Manazir* (optics) - Ibnal Haitham

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Free notes for KPK Board students, from A Textbook of Physics, Grade IX, KPTB Peshawar - Unit 1.